Gokul Raj

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SUMMARY

Embedded Systems Engineer with 2 years of experience in firmware development for IoT devices using C and C++ on ARM microcontrollers.

SKILLS

Programming: Embedded C, C++, Python, Assembly (basic)
Microcontrollers: STM32, ESP32, Arduino, Raspberry Pi
Protocols: UART, SPI, I2C, CAN, MQTT, Bluetooth Low Energy
Tools: Keil, STM32CubeIDE, FreeRTOS, Git, Oscilloscope, Logic Analyzer
Other: PCB design (KiCad), Linux, IoT

EXPERIENCE

Embedded Software Engineer, VoltEdge Electronics, Kinfra Hi-Tech Park Kalamassery | Aug 2024 - Present

- Wrote FreeRTOS firmware for a smart energy meter shipped to 8,000 homes.
- Implemented MQTT over Wi-Fi on ESP32 to send readings to the cloud every 15 minutes.
- Cut device power consumption by 22% using sleep modes.

Trainee Engineer, Keltron, Thiruvananthapuram | Jul 2023 - Jul 2024

- Tested and debugged CAN bus modules for electric vehicle chargers.

PROJECTS

Smart Irrigation Controller | ESP32, Soil Sensors, MQTT

- Automated watering for a 1-acre farm, saving 30% water.

EDUCATION

B.Tech in Electrical and Electronics Engineering, NSS College of Engineering, Palakkad | 2019 - 2023

- CGPA: 7.9 / 10